THE CHAIN RULE

DIFFERENTIAL CALCULUS

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TOPIC OUTLINE

The Chain Rule

Leibniz Notation

THE CHAIN RULE

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function defined by

$$F(x) = f(g(x))$$

is differentiable at x and F' is given by product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

Differentiate the function

$$F(x) = \sqrt{x^2 + 1}$$

Let $g(x) = x^2 + 1$ $\qquad f(g(x)) = \sqrt{g(x)}$

$$F'(x) = f'\left(\sqrt{g(x)}\right) \cdot g'(x^2 + 1)$$

$$F'(x) = \frac{1}{2}(g(x))^{-1/2} \cdot (2x)$$

$$F'(x) = \frac{x}{\sqrt{g(x)}}$$

$$\boxed{F'(x) = \frac{x}{\sqrt{x^2 + 1}}}$$

ans

LEIBNIZ NOTATION

If $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Differentiate the function

$$F(x) = \sqrt{x^2 + 1}$$

Let $u = x^2 + 1$ $\qquad f(u) = \sqrt{u}$

$$\frac{dF(x)}{dx} = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$\frac{dF(x)}{dx} = \frac{d}{du}\sqrt{u} \cdot \frac{d}{dx}\left(x^2 + 1\right)$$

$$\frac{dF(x)}{dx} = \frac{1}{2}u^{-1/2} \cdot 2x$$

$$\frac{dF(x)}{dx} = \frac{x}{\sqrt{u}}$$

$$\boxed{\frac{dF(x)}{dx} = \frac{x}{\sqrt{x^2 + 1}}}$$

ans

Differentiate $y = \sin x^2$

Let $u = x^2$ $\quad f(u) = \sin u$

$$\frac{dy}{dx} = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{d}{du}\sin u \cdot \frac{d}{dx}x^2$$

$$\frac{dy}{dx} = \cos u \cdot 2x$$

$$\boxed{\frac{dy}{dx} = 2x\cos x^2}$$

ans

Differentiate $y = x\sin\frac{1}{x}$

$$\frac{dy}{dx} = \left(\sin\frac{1}{x}\right)\frac{d}{dx}(x) + x\frac{d}{dx}\left(\sin\frac{1}{x}\right)$$

$$\frac{dy}{dx} = \sin\frac{1}{x}(1) + x\frac{d}{dx}\left(\sin\frac{1}{x}\right)$$

$$\frac{dy}{dx} = \sin\frac{1}{x} + x\left(-\frac{1}{x^2}\cos\frac{1}{x}\right)$$

$$\boxed{\frac{dy}{dx} = \sin\frac{1}{x} - \frac{1}{x}\cos\frac{1}{x}}$$

ans

$$F(x) = \sin\frac{1}{x}$$

Let $u = \frac{1}{x}$ $\quad f(u) = \sin u$

$$F'(x) = \frac{df(u)}{du}\cdot\frac{du}{dx}$$

$$F'(x) = \frac{d}{du}\sin u \cdot \frac{d}{dx}\left(\frac{1}{x}\right)$$

$$F'(x) = \cos u \cdot \left(-\frac{1}{x^2}\right)$$

$$F'(x) = -\frac{1}{x^2}\cos\frac{1}{x}$$

Differentiate $y = \frac{x^3}{\sqrt{1+x^2}}$

$$\frac{dy}{dx} = \frac{\sqrt{1+x^2}\,\frac{d}{dx}x^3 - x^3\,\frac{d}{dx}\sqrt{1+x^2}}{\left(\sqrt{1+x^2}\right)^2}$$

$$\frac{dy}{dx} = \frac{\left(\sqrt{1+x^2}\right)3x^2 - x^3\,\frac{d}{dx}\sqrt{1+x^2}}{1+x^2}$$

$$\frac{dy}{dx} = \frac{\left(\sqrt{1+x^2}\right)3x^2 - x^3\left(\frac{x}{\sqrt{1+x^2}}\right)}{1+x^2}$$

$$\frac{dy}{dx} = \frac{3x^2\sqrt{1+x^2} - \frac{x^4}{\sqrt{1+x^2}}}{1+x^2}$$

$$F(x) = \sqrt{1+x^2}$$

Let $u = 1 + x^2$ $\qquad f(u) = \sqrt{u}$

$$F'(x) = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$F'(x) = \frac{d}{du}\sqrt{u} \cdot \frac{d}{dx}\left(1+x^2\right)$$

$$F'(x) = \frac{1}{2}u^{-1/2} \cdot (2x)$$

$$F'(x) = \frac{x}{\sqrt{u}}$$

$$F'(x) = \frac{x}{\sqrt{1+x^2}}$$

Differentiate $y = \frac{x^3}{\sqrt{1+x^2}}$

$$\frac{dy}{dx} = \frac{\sqrt{1+x^2}\,\frac{d}{dx}x^3 - x^3\,\frac{d}{dx}\sqrt{1+x^2}}{\left(\sqrt{1+x^2}\right)^2}$$

$$\frac{dy}{dx} = \frac{\left(\sqrt{1+x^2}\right)3x^2 - x^3\,\frac{d}{dx}\sqrt{1+x^2}}{1+x^2}$$

$$\frac{dy}{dx} = \frac{\left(\sqrt{1+x^2}\right)3x^2 - x^3\left(\frac{x}{\sqrt{1+x^2}}\right)}{1+x^2}$$

$$\frac{dy}{dx} = \frac{3x^2\sqrt{1+x^2} - \frac{x^4}{\sqrt{1+x^2}}}{1+x^2}$$

$$\frac{dy}{dx} = \frac{3x^2\sqrt{1+x^2}\cdot\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}} - \frac{x^4}{\sqrt{1+x^2}}}{1+x^2}$$

$$\frac{dy}{dx} = \frac{\frac{3x^2(1+x^2)}{\sqrt{1+x^2}} - \frac{x^4}{\sqrt{1+x^2}}}{1+x^2}$$

$$\frac{dy}{dx} = \frac{3x^2+3x^4-x^4}{(1+x^2)\sqrt{1+x^2}}$$

$$\boxed{\frac{dy}{dx} = \frac{3x^2+2x^4}{(1+x^2)^{3/2}}}$$

ans

Differentiate $g(t) = \left(\frac{t-2}{2t+1}\right)^9$

Let $u = \frac{t-2}{2t+1}$ $\quad g(u) = u^9$

$$\frac{dg(t)}{dt} = \frac{dg(u)}{du} \cdot \frac{du}{dt}$$

$$\frac{dg(t)}{dt} = \frac{d}{du}u^9 \cdot \frac{d}{dt}\left(\frac{t-2}{2t+1}\right)$$

$$\frac{dg(t)}{dt} = 9u^8 \cdot \frac{(2t+1)\frac{d}{dt}(t-2) - (t-2)\frac{d}{dt}(2t+1)}{(2t+1)^2}$$

$$\frac{dg(t)}{dt} = 9u^8 \cdot \frac{(2t+1)(1) - (t-2)(2)}{(2t+1)^2}$$

$$\frac{dg(t)}{dt} = 9u^8 \cdot \frac{(2t+1) - (2t-4)}{(2t+1)^2}$$

$$\frac{dg(t)}{dt} = 9u^8 \cdot \frac{5}{(2t+1)^2}$$

$$\frac{dg(t)}{dt} = 9\left(\frac{t-2}{2t+1}\right)^8 \cdot \frac{5}{(2t+1)^2}$$

$$\boxed{\frac{dg(t)}{dt} = \frac{45(t-2)^8}{(2t+1)^{10}}}$$

ans

The displacement of a particle on a vibrating string is given by the equation

$$s(t) = 10 + \frac{1}{4}\sin(10\pi t)$$

where s is measured in centimeters and t in seconds. Find the **velocity** of the particle t seconds.

velocity

$$\frac{ds(t)}{dt} = \frac{1}{4}\frac{d}{dt}\sin(10\pi t)$$

Let $u = 10\pi t$ $\quad f(u) = \sin u$

$$\frac{ds(t)}{dt} = \frac{1}{4}\frac{df(u)}{du}\cdot\frac{du}{dt}$$

$$\frac{ds(t)}{dt} = \frac{1}{4}\frac{d}{du}\sin u\cdot\frac{d}{dt}10\pi t$$

$$\frac{ds(t)}{dt} = \frac{1}{4}\cos u\cdot(10\pi)$$

$$\boxed{\frac{ds(t)}{dt} = \frac{5\pi}{2}\cos(10\pi t)}$$

ans

SEATWORK